

# **Transfer Learning for Cross-Market Commodity Return Forecasting**

Leo Specks

Supervisor: Prof. Massimo Guidolin  
MSc Finance, Università Bocconi

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# 1 Test Chapter

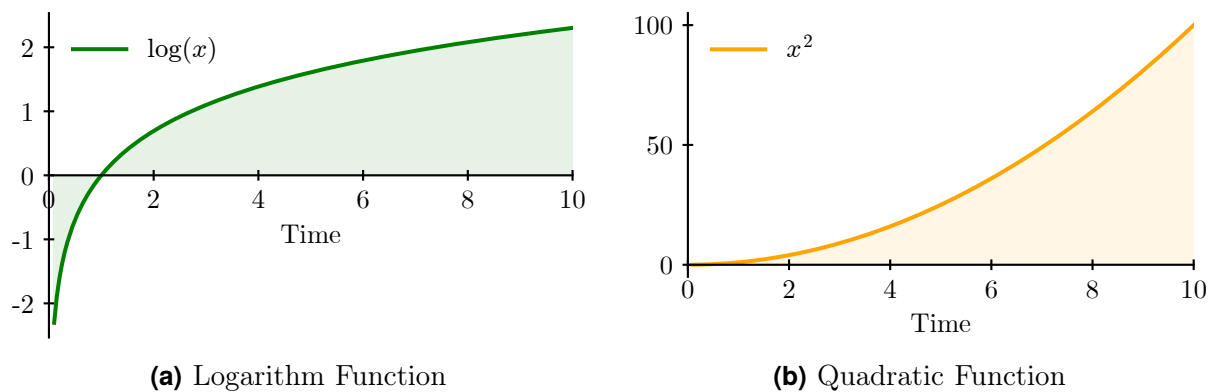
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**Figure 1**

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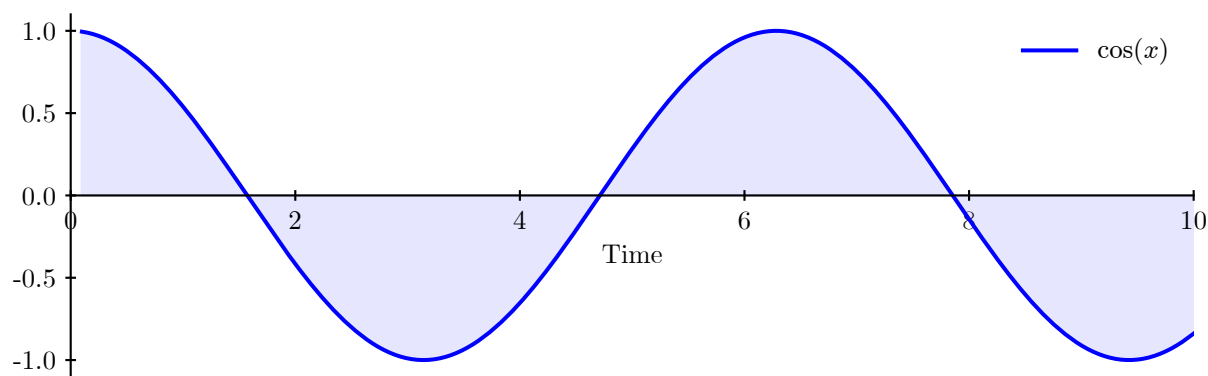


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**Figure 2**

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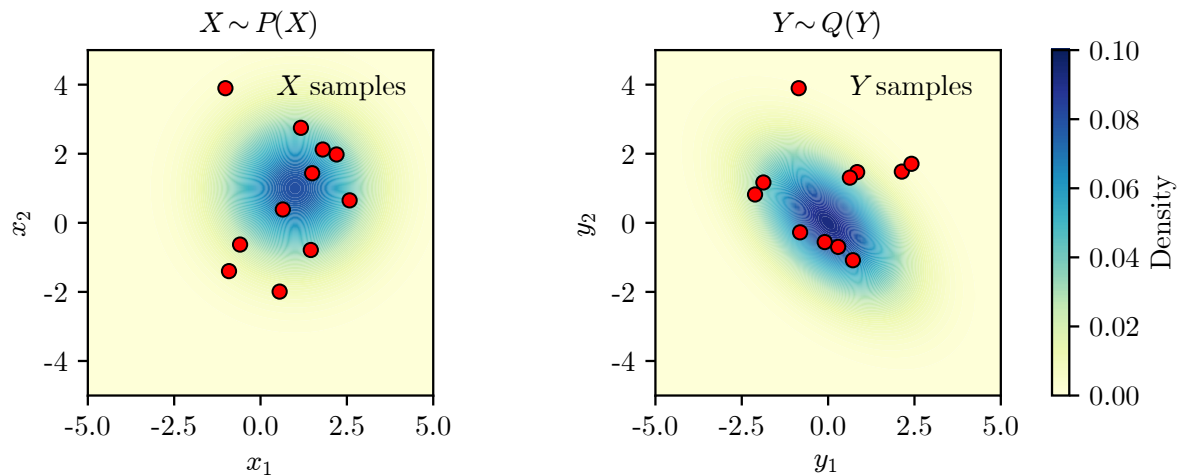
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**Figure 3**

MMD example plot



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## 1.2 Citations

This is a `\cite{}`: [2], this a `\textcite{}`: Guidolin and Pedio [2], this a `\parencite{}`: [2], this a `\citeauthor{}`: Guidolin and Pedio, this a `\citeyear{}`: 2021, and this a `\parencite` with two references: [1, 2].

**Table 1**  
Citation Styles with BibLaTeX

| Command                    | authoryear     | numeric           |
|----------------------------|----------------|-------------------|
| <code>\cite{}</code>       | Author, Year   | [index]           |
| <code>\textcite{}</code>   | Author (Year)  | Author [index]    |
| <code>\parencite{}</code>  | (Author, Year) | [index]           |
| <code>\citeauthor{}</code> | Author         | Author            |
| <code>\citeyear{}</code>   | Year           | Year (not linked) |

Use `\parencite{}` at the end of a statement and `\textcite{}` when referencing in the middle of a sentence.

## 1.3 Codeblock

The algorithm can be viewed in Codeblock 1. Nullam egestas viverra purus vitae varius. Sed maximus tempor vestibulum. Nunc tempor nulla justo, a posuere lorem tincidunt eget. Praesent dictum sed eros in pharetra. Vivamus nec dictum risus. Integer ac massa at magna rutrum pulvinar eget nec nunc. Curabitur tempor at lorem vitae venenatis. Suspendisse auctor posuere iaculis. Suspendisse egestas lobortis diam. Integer mattis.



**Code 1**

Example Codeblock

```
# KPI OVERVIEW
kpis = pd.DataFrame({
    "Ann. Return (approx)": ann_return_simple,
    "Ann. Return (compounding)": ann_return,
    "Ann. Volatility": ann_vol,
    "Sharpe": sharpe,
    "CAGR": cagr,
    "Max Drawdown": max_dd
})

kpis.T
```

## **2 Introduction**

### **2.1 Motivation & Background**

### **2.2 The Research Gap**

### **2.3 Research Questions & Hypotheses**

### **2.4 Contributions to the Literature**

### **2.5 Thesis Outline**

## 3 Literature Review

### 3.1 Theories of Commodity Futures Returns

### 3.2 Prevailing Commodity Forecasting Approaches

### 3.3 Cross-Market Transferability

Some of the more complex models seen in Subsection 3.2 suffer from needing a lot of data. For some more niche markets with shorter histories and more constrained datasets this raises a problem. The most straightforward and perhaps naive solution is to simply train models on data-rich markets and implement them on data-sparse ones. However, it has been shown that traditional models fail when being transferred cross-market. When trying to transfer a convolutional neural network (CNN) model trained on US commodity futures to predict returns of the Chinese market, Hao et al. [3] find that this model fails to systematically outperform models trained on data specific to the Chinese market. These results suggest that there are no significant return-predictive patterns from US commodity futures which generalize across different markets. Hao et al. [3] give two explanations on why employing models cross-market fails in commodity markets. **insert reasons** Furthermore, Nguyen and Walther [4] have also found results to be heterogeneous across different commodity futures as no single model fit all commodities. It seems that traditional models lack cross-market transferability across both different countries and commodities.

This shortcoming naturally invites the exploration of methodologies, which aim to build predictors, that allow for differences cross-market. The goal is to leverage data-rich markets, while still being able to include market-specific characteristics. This approach can be categorised as a transfer learning technique, which is a subset of machine learning, that allows for the assumption of different domains, tasks and distributions between the training and test set [5].

#### 3.3.1 Domain Adaptation and Fine-Tuning

Mashetty et al. [6] **explain approach** This approach proved to outperform both a traditional machine learning model and a deep learning model, which were trained only on target data, in all three chosen metrics, namely mean absolute error (MAE), root mean squared error (RMSE) and directional accuracy.

## 4 Methodology

### 4.1 Baseline Framework (Replication)

We begin by gathering all  $T$  observations of  $n$  variables into a data matrix  $\mathbf{X} \in \mathbb{R}^{T \times n}$ , where each row is a month and each column is a macroeconomic variable. Each column is demeaned and standardized to unit variance using the StandardScaler:

$$X_{\text{scaled},ij} = \frac{X_{\text{raw},ij} - \mu_j}{\sigma_j}. \quad (1)$$

Let  $\Sigma = V[\mathbf{X}] \in \mathbb{R}^{n \times n}$  denote the variance-covariance matrix of  $\mathbf{X}$ , which is calculated as:

$$\Sigma = \frac{1}{T-1} \mathbf{X}'\mathbf{X}, \quad (2)$$

and let  $\Sigma = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$  be its spectral decomposition, where  $\mathbf{U} \in \mathbb{R}^{n \times n}$  is an orthogonal matrix of order  $n$ , whose columns are the eigenvectors  $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$  of  $\Sigma$ , and  $\mathbf{\Lambda} \in \mathbb{R}^{n \times n}$  is a diagonal matrix of the same order, whose diagonal entries are the corresponding eigenvalues  $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$ , which, by convention, are sorted from the largest to the smallest. By multiplying the original scaled data matrix  $\mathbf{X}$  with  $\mathbf{U}$ , we derive the principal components of  $\mathbf{X}$  as the elements of the matrix  $\mathbf{F} \in \mathbb{R}^{T \times n}$ , or formally:

$$\mathbf{F} = \mathbf{X}\mathbf{U}. \quad (3)$$

To assess how much of the total variance in the original  $n$  variables is captured by the  $k$ -th principal component, or alternatively by the first  $K$  principal components, we can compute the variance-covariance matrix of  $\mathbf{F}$  as:

$$\begin{aligned} V[\mathbf{F}] &= V[\mathbf{X}\mathbf{U}] \\ &= \mathbf{U}'\Sigma\mathbf{U} \\ &= \mathbf{U}'\mathbf{U}\mathbf{\Lambda}\mathbf{U}'\mathbf{U} \\ &= \mathbf{I}_m\mathbf{\Lambda}\mathbf{I}_m \\ &= \mathbf{\Lambda}, \end{aligned} \quad (4)$$

where we exploited the orthogonality of  $\mathbf{U}$ . Recalling the fact that  $\mathbf{\Lambda}$  is a diagonal matrix,  $V[\mathbf{F}] = \mathbf{\Lambda}$  implies that the principal components are mutually uncorrelated, and have a variance equal to the associated eigenvalue. Therefore, the proportion of variance in the original  $n$  variables explained by the  $k$ -th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^n \lambda_i}, \quad (5)$$

and the cumulative proportion of explained variance by the first  $K$  components by:

$$\frac{\sum_{k=1}^K \lambda_k}{\sum_{i=1}^n \lambda_i}. \quad (6)$$

Mirroring the approach used by Guidolin and Pedio [2], we forecast the returns of each commodity  $j$ ,  $R_{t+1,j}$ , with the principal component analysis to summarize a large set of macroeconomic factors:

$$R_{t+1,j} = \underbrace{\alpha_j}_{\text{intercept}} + \underbrace{\sum_{i=1}^N \beta_{ij} PC_{t,ij}}_{\text{macroeconomic}} + \underbrace{D_{t,j} \left( \sum_{k=1}^K \gamma_{kj} C_{t,kj} \right)}_{\text{commodity-specific}} + \underbrace{\varepsilon_{t,j}}_{\text{error term}} \quad (7)$$

where  $\alpha_j$  is the model's intercept,

## 4.2 Extension I: Penalized Regression

In the first alternate approach, we keep the principal component analysis and still extract ten principal components from the  $n$  initial variables, but instead of running stepwise AIC on those selected principal components, we run a LASSO regression as an alternative way to further reduce the number of parameters to be estimated. With this only being a narrow replacement compared to the original framework, every difference in results should be attributable solely to the selection method.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{10} \beta_{ij}^{\text{LASSO}} PC_{t,ij} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (8)$$

A broader extension to the approach by Guidolin and Pedio [2] is foregoing the principal component analysis entirely and applying LASSO directly to the  $n$  variables.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{128} \beta_{ij}^{\text{LASSO}} X_{t,i} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (9)$$

## 4.3 Extension II: Individual Fundamental Predictors

### 4.4 OOS Evaluation Criteria

### 4.5 Portfolio Back-Testing Framework

## **5 Data**

### **5.1 Commodity Futures Returns**

### **5.2 Macroeconomic Factors**

### **5.3 Aggregate Commodity-Specific Factors**

### **5.4 Individual Fundamental Predictors**

### **5.5 Sample Splits and Descriptive Statistics**

## **6 Replication & Baseline Results**

### **6.1 In-Sample Estimation (1989-2003)**

### **6.2 OOS Predictive Accuracy (2004-2012)**

### **6.3 Portfolio Back-Testing (2004-2012)**

### **6.4 Replication Assessment**

## **7 Extension I: ML Variable Selection**

### **7.1 Model Specification and Implementation**

### **7.2 Variable Selection Patterns**

### **7.3 OOS Forecasting Results - Original Sample (2004-2012)**

### **7.4 OOS Forecasting Results - Extended Sample (2013-2024)**

### **7.5 Portfolio Back-Testing**

### **7.6 Robustness Checks**



## **8 Extension II: Individual Commodity Fundamentals**

### **8.1 Predictor Construction by Commodity Group**

### **8.2 Individual vs. Aggregate Factors: Are They Substitutes?**

### **8.3 OOS Forecasting Results (2004-2024)**

### **8.4 Combined Model**

### **8.5 Portfolio Back-Testing**

### **8.6 Regime Interaction**

## **9 Synthesis & Discussion**

### **9.1 Model Horse Race**

### **9.2 Statistical vs. Economic Loss Functions Revisited**

### **9.3 Implications for Commodity Market Segmentation**

### **9.4 Practical Implications**

## **10 Conclusions**

### **10.1 Summary of Findings**

### **10.2 Limitations**

### **10.3 Future Research Directions**

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## List of Acronyms

**CNN** Convolutional Neural Network

**MAE** Mean Absolute Error

**RMSE** Root Mean Squared Error